MAIS PAC

IPhO 02.01: Kinematics: Describing Motion · Worked Solutions

Kinematics: Describing Motion: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

Solution index

1 2 3 4 5 6 7 8 9 10

Solution 1 — From a v–t graph to real motion. Return →

(a) Stage 1 area 4.0×2.0 = +8.0 {m}. Stage 2 is a trapezium with average velocity (4.0 + (-2.0))/2 = 1.0 {m,s^(-1)} over 3.0 {s}: +3.0 {m}. Displacement = +11.0 {m}.

(b) In stage 2 the velocity crosses zero after travelling 4.0/(6.0/3.0)=2.0 {s}, so the forward part has area +(1)/(2)(4.0)(2.0)=+4.0 {m} and the reverse part -(1)/(2)(2.0)(1.0) = -1.0 {m}. Total distance = 8.0 + 4.0 + 1.0 = 13.0 {m} ≠ 11.0 {m} in magnitude of displacement. **Check:** distance ≥ |displacement|, as required.

(c) a = (-2.0-4.0)/3.0 = -2.0 {m,s^(-2)}. The turn-around occurs when v=0, at t = 2.0 + 4.0/2.0 = 4.0 {s} after the start.

Solution 2 — Two stones, one clock. Return →

(a) Let t be measured from the release of the first stone. v_1 = gt; the second has v_2 = 12 + g(t-1) for t 1. Setting them equal: gt = 12 + g(t-1) 0 = 12 - g, which is false. The second stone is **always** faster by 12 - g = 2.2 {m,s^(-1)} while both are in flight — a relative-motion statement that needs no algebra once seen.

(b) The speed difference is constant, so there is no instant of equal speed; the question is a trap and the correct answer is that they never share a speed while both fall freely.

(c) The second stone starts 1.0 {s} later and 12 {m,s^(-1)} faster, so its displacement always differs from the first by the initial gap (1)/(2) g(1.0)² plus the accumulated lead 2.2t; both terms are positive, so positions never coincide. **Check:** the limiting case 12 = g gives equal accelerations and equal speeds only if also equal start times — consistent.

Solution 3 — Range on level ground and on a wall. Return →

(a) R_(max) = v_0²/g = 400/9.8 = 40.8 {m} at 45°.

(b) 25 = (400sin 2θ)/(9.8) sin 2θ = 0.6125 2θ = 37.8° or 142.2°, so θ ≈ 18.9° or 71.1° — complementary, as expected.

(c) Trajectory: y = xtan θ - (9.8x²)/(2(400)cos ^2θ). At x = 20 {m}: 6.0 < 20tan θ - 4.9cos ^{-2}θ. Writing c=cos ^2θ and tan θ = √((1-c)/c) and solving numerically gives 32° θ 70°. **Check:** θ = 45° gives y = 20 - 4.9 = 15.1 {m}, comfortably above 6 {m}, while θ = 10° gives only about 3.5 {m} — the window is sensible.

Solution 4 — The river crossing. Return →

(a) Aim upstream at angle θ to the bank with 1.6sin θ = 1.2 θ = 48.6°. Cross-stream speed 1.6cos θ = 1.06 {m,s^(-1)}, so t = 80/1.06 = 75 {s}.

(b) Aim straight across: cross-stream speed 1.6 {m,s^(-1)}, t = 50 {s}, drifting 1.2×50 = 60 {m} downstream.

(c) A straight crossing needs v_({row})sin θ = u with sin θ1, impossible when u > v_({row}). At u = v_({row}) the only solution is θ = 90°, i.e. aiming straight upstream with zero cross-stream progress — the crossing takes infinite time. **Check:** the fast-crossing time 50 {s} is shorter than the straight-crossing time 75 {s}, which is why ferry timetables care about the current.

Solution 5 — Projectile over a wall. Return →

(a) The height at x=d is y(d) = dtan θ - (gd²)/(2v_0²cos ^2θ). Requiring y(d) H gives

$$v_0^2 \ (gd^2)/(2\cos ^2\theta,(d\tan \theta - H)). \qquad (1)$$

(b) Minimising the required v_0² over θ (differentiate the denominator) gives tan θ = (d)/(d - 2H), the **envelope** result: the optimal launch aims above the target, because the extra height is bought cheaply.

(c) The standard maximum-range-envelope condition is v_0² g(y+√(x²+y²)); the square-root term is the straight-line distance from the launch point to the target, so the condition says the **reachable** boundary is an upward-opening paraboloid of revolution. **Check:** for y=0 it reduces to v_0² gx, i.e. x v_0²/g = R_(max), which is exactly right.

Solution 6 — Reading an acceleration graph. Return →

(a) a(t) is three horizontal segments: +2, 0, -3. v(t) rises linearly to a peak, stays flat, then falls linearly to zero — a trapezium. x(t) has increasing positive slope, then constant slope, then decreasing slope flattening out — no inflection is lost, but its curvature changes sign.

(b) v_(max) = 2.0×3.0 = 6.0 {m,s^(-1)} at t = 3.0 {s}, held until t = 5.0 {s}.

(c) The final stage lasts 6.0/3.0 = 2.0 {s}, so the total is 7.0 {s}. Position: (1)/(2)(2)(9) = 9.0 {m} in stage 1, 6.0×2.0 = 12.0 {m} in stage 2, and (1)/(2)(6.0)(2.0) = 6.0 {m} in stage 3: x_({final}) = 27.0 {m}. **Check:** total area under v(t) equals the sum of the three areas, and the object never reverses, so displacement equals distance.

Solution 7 — Drag changes everything. Return →

(a) Terminal speed is reached when drag balances weight: mg = kv_T v_T = mg/k. The equation m v = mg - kv can be written v = (v_T - v)/τ with τ = m/k.

(b) Solving with v(0)=0 gives v = v_T(1-e^(-t/τ)), and integrating, y = v_Tt - v_Tτ(1-e^(-t/τ)).

(c) At t = 3τ, e^(-3) = 0.0498, so y = v_Tτ(3 - 0.950) = 2.05,v_Tτ while v_Tt = 3v_Tτ: only 68% of the drag-free distance. **Check:** as t→∞, y→ v_T t - v_Tτ, i.e. the drop lags the terminal-speed line by a constant v_Tτ — the classic drag “offset”.

Solution 8 — Two-dimensional chase. Return →

(a) The relative position is (3t, 40 - 5t), so D(t)=√(9t² + (40-5t)²).

(b) Minimise D² = 34t² - 400t + 1600: t = 400/68 = 5.88 {s}, and D_(min) = √(1600 - 400²/(4·34)) = √(1600-1176) = 20.6 {m}.

(c) At t = 5.88 {s} the second boat is at (0, 40 - 29.4) = (0, 10.6) {m}. A boat leaving the origin must therefore have velocity (0, 10.6/5.88) = (0, 1.80) {m,s^(-1)}. **Check:** the answer follows from the relative-velocity picture — the minimum-separation geometry is a right angle between the relative velocity and the line of sight.

Solution 9 — Throwing from a cliff. Return →

(a) -h = v_0sin θ, t - (1)/(2) gt², so (1)/(2)gt² - v_0sin θ,t - h = 0 and

$$t = (v_0 \sin\theta + \sqrt{(v_0^2 \sin^2\theta + 2gh)})/(g). \qquad (2)$$

The negative root would place the launch in the past, so it is discarded on physical grounds.

(b) Substituting into R = v_0cos θ,t gives the stated expression.

(c) For large h the square root is dominated by √(2gh), independent of θ, so R ≈ v_0cos θ√(2h/g), which is greatest at θ→0: from a very high cliff you should throw horizontally and let gravity do the work. **Check:** for h=0 the formula collapses to R = v_0²sin 2θ/g, as it must.

Solution 10 — A rotating reference frame. Return →

(a) In the rotating frame the radial acceleration contains the centrifugal term +ω²r and the Coriolis term 2ωθ; because the groove constrains θ = 0 for the bead in the groove, only the centrifugal term survives.

(b) Therefore r = ω² r, with solution r = Aω t + Bω t. With r(0)=r_0 and r(0)=0, r = r_0ω t.

(c) In the inertial frame the bead has tangential speed ω r_0 and, with no tangential force, moves in a straight line; the groove curves away from it and must push the bead inward, so the required sideways force grows as ω² r. The “centrifugal force” is the bookkeeping of that constraint.

(d) For ω t ≪ 1, ω t ≈ 1 + (1)/(2)ω²t², so r ≈ 0: over short times the bead barely notices the rotation, in agreement with the inertial-frame picture. **Check:** energy in the inertial frame, (1)/(2)mω²r², is conserved; in the rotating frame the same expression appears as the centrifugal potential.